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Passkeys in Practice: An Empirical Evaluation of FIDO2/WebAuthn Compliance and Interoperability

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***Abstract*—Passwordless authentication is rapidly reshaping digital identity management, with passkeys emerging as a key technology prepared to replace traditional passwords. Built on the FIDO2 and WebAuthn standards, passkeys promise secure, phishing-resistant, and user-friendly authentication. Despite increasing industry adoption, their real-world interoperability across browsers and devices remains untested.**

This paper presents one of the first empirical studies evaluating passkey implementations in Chrome, Safari, Edge, and Firefox across four passkey types: iCloud Keychain, browser profile credentials, phone or tablet–based QR code workflows, and USB security keys. We propose a structured evaluation framework and introduce metrics to assess cross-browser credential reuse, cross-device synchronization, and roaming authenticator interoperability. Our results expose cross-browser inconsistencies driven by non-uniform credential management, consent handling, and error reporting. The study contributes a structured evaluation framework, reproducible criteria, and interoperability metrics that benchmark passkey readiness across major browsers.

***Index Terms*—Passkeys, passwordless authentication, FIDO2, WebAuthn, interoperability, compliance, identity management.**

I. Introduction

Authentication is a fundamental component of secure Internet systems, acting as the first line of defense against unauthorized access and identity-based attacks. Traditional password-based authentication has long been criticized for its inherent weaknesses [1]. Human-chosen passwords are prone to reuse, guessability, and phishing, leading to widespread breaches and escalating security incidents [2], [3]. To overcome these shortcomings, passwordless authentication—commonly referred to as passkeys—has emerged as a promising paradigm [4], leveraging public-key cryptography, biometrics, and device-bound credentials to eliminate shared secrets and reduce the attack surface [4].

Despite growing industry adoption, passkey implementations face significant interoperability and compliance challenges [5]–[7]. Variations in platform support (e.g., iOS, Android, Windows), inconsistent adherence to FIDO2 and WebAuthn standards, and vendor-specific implementations have led to fragmentation. Such inconsistencies hinder the goal of seamless cross-platform authentication, creating usability issues and potential security risks [8]. Addressing these gaps is essential for achieving universal, secure, and user-friendly passwordless authentication.

Ensuring that passkey implementations comply with FIDO2/WebAuthn specifications is critical to fostering trust and widespread adoption [9]. Non-compliant or partially implemented features can weaken cryptographic assurances, reduce usability, and disrupt cross-browser and cross-device interoperability. A systematic evaluation of real-world deployments is therefore necessary to identify deviations and provide actionable insights for standardization and implementation improvements [10].

This paper presents an empirical evaluation of passkey implementations across four major browsers—Chrome, Safari, Edge, and Firefox—using multiple passkey types: iCloud Keychain, browser profile credentials, phone–based QR code workflows, and USB security keys. Four experimental scenarios were considered: initial registration, duplicate registration, cross-browser authentication, and cross-device authentication. The study assesses both compliance with FIDO2/WebAuthn specifications and practical interoperability.

The results reveal that while iCloud Keychain and USB security keys achieved full cross-browser reuse, browser profile credentials failed entirely outside their originating browser. QR code–based workflows exhibited inconsistent support, particularly in Firefox and Safari. Cross-device synchronization was unreliable, with Chrome, Edge, and Firefox succeeding in half of the attempts, while Safari performed the worst. Safari also lacked support for roaming authenticators. These findings underscore that specification compliance alone does not ensure seamless interoperability, highlighting the need for standardized fallback mechanisms, consistent QR-based workflows, and improved error handling. This paper makes the following key contributions:

- A reproducible framework mapping WebAuthn/CTAP2 specifications to practical compliance criteria.
- A cross-browser empirical study of passkeys using iCloud Keychain, browser profiles, QR code workflows, and USB security keys.
- Identification of key inconsistencies in compliance and interoperability, with recommendations to enhance usability and adoption.

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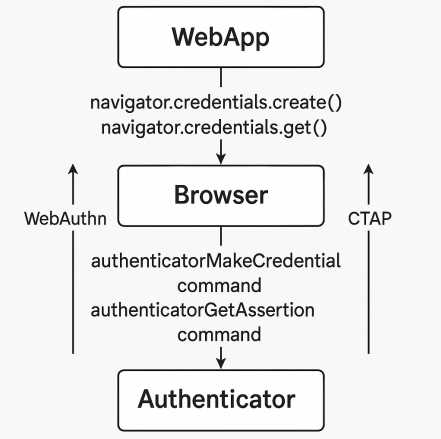


Fig. 1: The flowchart illustrates the interaction between a Web Application, the Browser, and an Authenticator within the FIDO2 authentication framework.

II. Background and Related Work

Passwordless authentication has emerged as a promising alternative to traditional passwords, addressing usability and security challenges such as password reuse, phishing, and credential stuffing attacks [11]. Central to this paradigm are the FIDO2 and WebAuthn standards, which provide strong cryptographic guarantees and improved user experience through a device-bound or synchronized passkey.

A. Web Authentication (WebAuthn)

The Web Authentication (WebAuthn) specification, standardized by the World Wide Web Consortium (W3C), defines a browser API that enables web applications to register and authenticate users using public-key cryptography. As shown in Figure 1, WebAuthn provides high-level methods, such as `navigator.credentials.create()` for credential registration and `navigator.credentials.get()` for authentication. These API calls are handled by the browser, which mediates communication with authenticators while enforcing origin scoping, user consent, and privacy-preserving operations.

B. Client to Authenticator Protocol (CTAP)

The Client to Authenticator Protocol (CTAP), specified by the FIDO Alliance, complements WebAuthn by defining how browsers and operating systems communicate with authenticators. As shown in Figure 1, the browser translates WebAuthn API calls into CTAP commands, such as `authenticatorMakeCredential` for credential creation and `authenticatorGetAssertion` for authentication. CTAP supports multiple transport mechanisms (USB, NFC, Bluetooth Low Energy) and exists in two versions: CTAP1 (U2F authenticators) and CTAP2 (FIDO2 authenticators with advanced capabilities).

This separation of WebAuthn (application-facing API) and CTAP (authenticator communication) enables modular and secure passwordless authentication. However, deviations in CTAP implementation or browser handling of WebAuthn requests can lead to interoperability issues, motivating our evaluation of cross-browser and cross-device passkey behavior.

C. Prior Research on Authentication Compliance

Prior studies have explored interoperability and compliance issues within FIDO2/WebAuthn ecosystems. For instance, researchers have highlighted the challenges posed by platform-specific extensions, version drift, and inconsistent implementations across browsers and operating systems [5], [7], [12]. Other works have focused on enhancing privacy-preserving attestation mechanisms and flexible authenticator architectures to improve system compatibility [13], [14]. However, few evaluations have empirically assessed real-world passkey interoperability across multiple browsers and passkeys [15]–[17]. While prior research has primarily focused on theoretical security properties, protocol design improvements, or compliance testing frameworks, there has been limited empirical evaluation of passkey interoperability in real-world browser environments. Our study differs by conducting a comprehensive, scenario-based assessment of passkey behavior across four major browsers (Chrome, Safari, Edge, and Firefox) and multiple passkey types, including iCloud Keychain, browser profile credentials, phone or tablet–based QR code workflows, and USB security keys. By combining compliance analysis with practical interoperability measurements, our work provides actionable insights into implementation gaps that directly affect user experience and the broader adoption of passwordless authentication.

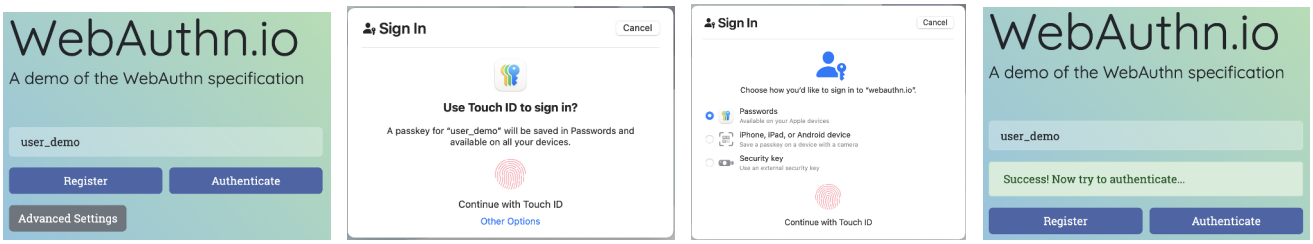
III. Methodology

This study adopts an empirical approach to evaluate whether major web browsers comply with the FIDO2/WebAuthn specifications for passkey registration and authentication workflows. It further investigates their ability to support seamless cross-browser and cross-device interoperability. The evaluation emphasizes observable user interface behaviors, thereby reflecting the practical experience of end-users interacting with passkeys across different devices and authenticators.

A. Study Objective and scope

The primary objective of this work is to assess the extent to which current browser implementations adhere to the FIDO2/WebAuthn standards while supporting practical interoperability. Although these standards have undergone rigorous technical validation, previous studies have shown that compliance alone does not guarantee seamless cross-platform usability [5], [7]. Building on this insight, we evaluate the behavior of four widely used browsers—Chrome, Safari, Edge, and Firefox—when employing four passkey types: iCloud Keychain, browser profile credentials, phone or tablet–based QR code workflows, and USB security keys. The evaluation

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(a) Registration Screen. (b) Default passkey. (c) Passkeys list. (d) Notification

Fig. 2: Example screenshots showing the process of registration.

focuses on both specification compliance and practical interoperability, providing a holistic perspective on the current state of passkey deployment.

B. Experimental Setup

Experiments were conducted on an Apple MacBook Pro (M3 Max) running macOS Sequoia 15.6.1, using the latest stable releases of Chrome 141.0.7390.77, Safari 18.6, Edge 141.0.3537.71, and Firefox 142.0. The WebAuthn.io testbed was used as the relying party with its default configuration (`webauthn.io`), ensuring standard WebAuthn behavior without custom server-side changes.

Four passkey types were tested: (i) platform-synced credentials via iCloud Keychain, (ii) browser profile credentials, (iii) cross-device authentication via QR codes, and (iv) roaming authenticators using a YubiKey 5C NFC USB device. Each registration and authentication flow was executed multiple times under identical network and system conditions. Screenshots and console logs were captured to record outcomes, error messages, and fallback behaviors, enabling reproducibility across all test cases.

C. Data Collection

Data collection focused on capturing the observable behaviors of each browser and passkey type during registration and authentication attempts. For every test attempt, the browser used, the passkey type (iCloud Keychain, browser profile credentials, phone or tablet–based QR code workflows, or USB security keys), and the outcome (success or failure) were recorded. Any error messages displayed in the user interface or shown in browser developer tools were documented in detail.

To provide visual evidence of the observed workflows, representative screenshots were captured for all experimental scenarios. These screenshots illustrate browser prompts, credential selection dialogs, and final success or error messages. Figure 2 shows an example of the registration process using the WebAuthn.io testbed, including the registration screen, passkey confirmation prompt, list of available passkeys, and success notification.

D. Experimental Scenarios

The experimental framework was structured around four principal scenarios designed to systematically evaluate both the technical implementation and user experience of passkey workflows across multiple browsers. These scenarios were carefully selected to provide comprehensive coverage of typical use cases while specifically targeting potential interoperability challenges and security considerations inherent in cross-browser passkey implementations.

To evaluate the compliance of browser implementations with FIDO2/WebAuthn specifications, we designed a set of experimental scenarios. For each scenario, we derived a set of observable compliance criteria based on the W3C WebAuthn Level 2 specification and FIDO2 Client-to-Authenticator Protocol (CTAP2). These criteria reflect the expected behaviors during passkey registration and authentication workflows as experienced through browser user interfaces.

1) **Initial Registration:** This scenario evaluated the process of creating a new passkey for a previously unregistered user account. Tests were performed across all browsers and passkey types, with attention to user interface prompts, consent dialogs, and error handling during first-time credential creation.
2) **Duplicate Registration:** This scenario examined how browsers handle attempts to register the same user account multiple times. It assessed whether browsers enforced exclusion lists and provided appropriate warnings or options when duplicate credentials were detected.
3) **Cross-Browser Authentication:** This scenario measured the ability to authenticate in a browser different from the one where the passkey was initially registered. Both successful and failed authentication attempts were recorded to evaluate practical credential portability and interoperability.
4) **QR Code-Specific Tests:** This scenario focused on QR code–based workflows. Tests were performed for both initial credential creation using a secondary device and subsequent authentication, reflecting the growing role of cross-device passkey usage [18].

Collectively, these scenarios provide broad coverage of both normative and edge cases, enabling the measurement of compliance and practical interoperability across major browsers and passkey types. This study focuses on observable UI-level behavior rather than packet-level CTAP tracing; future work will instrument browsers to capture low-level commands for

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TABLE I: Mapping Experimental Scenarios to Compliance Criteria. Each compliance criterion (C1–C4) was derived directly from the normative statements in the W3C WebAuthn Level 2 and FIDO2 CTAP2 specifications.

Scenario	Applicable Compliance Criteria
Initial Registration	C1.1 Browser prompts user to create a passkey (UI prompt visible). C1.2 All passkey types (platform, roaming, synced) offered. C1.3 User verification (PIN, biometric) enforced. C1.4 Consent prompt for credential creation displayed. C1.5 Registration success is clearly indicated in the UI. C1.6 Errors for unsupported authenticators are handled gracefully.
Duplicate Registration	C2.1 Browser checks for existing credentials and enforces exclusion lists. C2.2 UI shows a clear error or warning when duplicates are attempted. C2.3 Consistency of behavior across browsers.
Authentication	C3.1 Browser discovers synced or roaming credentials registered elsewhere. C3.2 UI prompts the user with available credentials for authentication. C3.3 Authentication succeeds without re-registration. C3.4 Failure scenarios (e.g., missing credentials) produce clear error messages. C3.5 Synced passkeys function across the ecosystem C3.6 Fallback Mechanisms
QR Code-Specific Tests	C4.1 QR code workflows are available for registration. C4.2 QR code workflows are available for authentication. C4.3 Cross-device QR handling functions correctly. C4.4 UI fallback options are presented if QR workflows are unsupported. C4.5 QR registration and authentication flows are consistent across browsers.

deeper protocol validation.

IV. Analysis Framework

The analysis framework evaluates browser behavior from two complementary perspectives: formal compliance with FIDO2/WebAuthn specifications and practical interoperability across browsers, devices, and passkey types. This dual approach ensures that the study not only verifies adherence to standards but also captures real-world usability implications.

A. Compliance Assessment

A systematic compliance assessment framework was adopted to evaluate passkey registration and authentication across browsers. Observable user interface behaviors were used as primary evidence of conformance. The four experimental scenarios-initial registration, duplicate registration, cross-browser authentication, and QR code–specific tests-were each associated with compliance criteria derived from the WebAuthn specification and FIDO2 CTAP2 protocol.

Table I summarizes the mapping between scenarios and their corresponding compliance criteria. Each criterion (C1–C4) was derived directly from normative clauses in the W3C WebAuthn Level 2 and FIDO CTAP2 specifications. For the *initial registration scenario*, criteria C1.1–C1.6 evaluate whether the browser displays a registration prompt, offers all supported passkey types, enforces user verification, requests user consent, clearly indicates success, and gracefully handles unsupported authenticators. For the *duplicate registration scenario*, criteria C2.1–C2.3 assess whether the browser checks for existing credentials, provides appropriate error messages for duplicates, and maintains consistent behavior across browsers. For the *authentication scenario*, criteria C3.1–C3.6 examine the discoverability of credentials created elsewhere, the presence of user prompts, the ability to authenticate without re-registration, the clarity of error messages, the correct functioning of synced passkeys, and the availability of fallback mechanisms. Finally, for the *QR code–specific tests*, criteria C4.1–C4.5 verify support for QR-based registration and authentication, correct handling of cross-device QR workflows, the presence of fallback options, and consistency of QR workflows across browsers. This structured mapping provides a foundation for systematically documenting the observed behaviors of each browser, identifying deviations from expected behaviors, and assessing overall compliance with FIDO2 and WebAuthn specifications.

B. Interoperability Evaluation

Interoperability is a key goal of the FIDO2 and WebAuthn standards, particularly through support for multi-device credentials and discoverable keys. However, implementation differences can hinder seamless passkey reuse across browsers and devices. To systematically assess practical interoperability across browsers, platforms, and passkey types, we define four focused metrics. These metrics capture critical aspects of cross-browser credential usability, device synchronization, roaming authenticator compatibility [19], and QR code-based fallback workflows. The following equations formalize each metric:

1) Cross-Browser Credential Reuse Rate (CBCRR): CBCRR measures how often a passkey created in one browser can be reused for authentication in other browsers on the same platform. A high CBCRR indicates that passkey implementations support seamless cross-browser authentication workflows without requiring redundant credential registration:

$$\text{CBCRR} = \frac{\text{Successful cross-browser authentications}}{\text{Total cross-browser authentications}}$$

2) Cross-Device Sync Success Rate (CDSSR): CDSSR evaluates whether passkeys, particularly those managed by services such as iCloud Keychain or Google Password Manager, synchronize properly across devices within the same ecosystem.

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This metric assesses whether a passkey created on one device is correctly propagated and can be used for authentication on a second device. CDSSR serves as a key indicator of ecosystem-wide credential portability and supports the WebAuthn goal of multi-device credentials:

$$\text{CDSSR} = \frac{\text{Successful cross-device authentications}}{\text{Total cross-device authentications}}$$

3) Roaming Authenticator Reuse Rate (RARR): RARR measures the interoperability of passkeys stored on external, hardware-based authenticators (e.g., USB security keys conforming to CTAP2). This metric captures whether such credentials can be used consistently across different browsers and platforms, reflecting adherence to the roaming authenticator specifications outlined in FIDO2. A high RARR indicates that browser implementations can successfully discover and interact with resident credentials through CTAP2-compliant devices, reinforcing device independence.

$$\text{RARR} = \frac{\text{Successful roaming authentications}}{\text{Total roaming authentications}}$$

These metrics provide a comprehensive view of practical interoperability by evaluating credential portability, ecosystem-wide synchronization, roaming authenticator usability, and support for cross-device QR workflows.

V. Results and Analysis

The results of this study are presented following the analysis framework outlined in Section 4. Each experimental scenario is evaluated against its associated compliance criteria (Table I), providing insights into browser behaviors, specification adherence, and practical interoperability. Findings are organized under the two core dimensions of the framework: Compliance Assessment and Interoperability Evaluation.

A. Compliance Assessment Results

Table II summarizes the compliance outcomes for all browsers across the four passkey types. Results are grouped according to the four experimental scenarios (i.e., section III-D). The compliance levels are represented as Pass, Partial, or Fail, indicating full adherence, partial adherence (e.g., incomplete UI prompts or inconsistent behavior), or non-adherence to the WebAuthn and CTAP2 specifications, respectively.

1) Initial Registration (C1.1–C1.6): Most browsers successfully met basic requirements for displaying passkey prompts (C1.1) and enforcing user verification (C1.3). However, all browsers failed to consistently display consent prompts (C1.4), an essential safeguard against silent credential creation. Partial compliance was observed for authenticator options (C1.2), especially for QR-based and browser profile credentials, indicating inconsistent availability of all passkey types during registration. Chrome and Safari demonstrated the most reliable registration success (C1.5), while Firefox frequently lacked proper error handling for unsupported authenticators (C1.6).

2) Duplicate Registration (C2.1–C2.3): Enforcement of exclusion lists (C2.1) failed across all browsers, with no exclusion enforcement occurring. Chrome and Safari correctly provided error messages when duplicates were attempted (C2.2), whereas Edge and Firefox displayed inconsistent behavior. None of the browsers achieved full consistency of behavior across browsers (C2.3).

3) Authentication (C3.1–C3.6): Chrome, Safari, and Edge generally performed well in credential discovery (C3.1) and authentication without re-registration (C3.3). Firefox showed failures with browser profile and QR-based credentials. UI prompts for credential selection (C3.2) were missing in some Firefox cases. Clear error messages for failed authentication (C3.4) were inconsistently presented, and fallback mechanisms (C3.6) were only partially supported. Safari and Chrome offered better cross-device fallback options than Edge and Firefox.

4) QR Code-Specific Tests (C4.1–C4.5): Safari and Edge demonstrated full support for QR-based registration and authentication criteria, while Safari and Firefox provided partial support. Both failed to present fallback options when QR workflows were unsupported (C4.4), and frequently showed inconsistent QR registration/authentication flows (C4.5).

B. Interoperability Evaluation Results

Figure 3 illustrates the CBCRR across major browsers (Chrome, Edge, Firefox, Safari) for different passkey types (iCloud Keychain, Browser Profile, QR Code, USB Security Key). CBCRR quantifies the ability to reuse a passkey created in one browser for authentication in other browsers on the same device.

As we can see, iCloud Keychain and USB Security Keys achieved full interoperability (CBCRR = 1.0) across all browsers, indicating that credentials created with these passkey types were consistently reusable in cross-browser scenarios. Browser Profile passkeys exhibited poor interoperability, with Chrome and Edge achieving only CBCRR = 0.25, while Firefox and Safari failed (CBCRR = 0.00). This highlights that browser profile-based credentials are not portable across browsers. We can consider this as the profile is restricted to the browser itself, and there is no sync between browsers for sharing accounts.

QR Code-based passkeys showed partial interoperability, with Chrome and Edge achieving full reuse (CBCRR = 1.0), but Firefox and Safari demonstrated only CBCRR = 0.50, indicating inconsistent credential discovery across browsers. These findings suggest that while platform-synced passkeys (iCloud Keychain) and roaming authenticators (USB Security Key) meet the goal of cross-browser usability, browser-profile credentials are highly sensitive, and QR-based credentials require improved standardization for consistent cross-browser behavior.

Figure 4 presents the CDSSR across Chrome, Edge, Firefox, and Safari, representing the proportion of successful authentications using synced passkeys across devices within the same

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TABLE II: Compliance assessment heatmap for passkey registration and authentication results across browsers and passkeys.

Scenario	Passkey Type	iCloud Keychain				Browser Profile				Phone/Tablet (QR Code)				USB Security Key			
	Browser	Chrome	Edge	Firefox	Safari	Chrome	Edge	Firefox	Safari	Chrome	Edge	Firefox	Safari	Chrome	Edge	Firefox	Safari
Registration	C1.1	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
	C1.2	Partial	Partial	Partial	Partial	Partial	Partial	Fail	Fail	Partial	Partial	Partial	Partial	Partial	Partial	Partial	Partial
	C1.3	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
	C1.4	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail
	C1.5	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
	C1.6	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Fail	Fail	Fail	Pass	Pass	Pass	Pass	Pass
Duplicate Registration	C2.1	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail	Fail
	C2.2	Pass	pass	Partial	Pass	Pass	Fail	Fail	Fail	Pass	Partial	Fail	Pass	Pass	Fail	Fail	Pass
	C2.3	Partial	Partial	Partial	Partial	pass	Pass	Fail	Fail	Fail	Fail	Fail	Pass	Pass	Pass	Partial	Pass
Authentication	C3.1	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Fail	Pass	Pass	pass	Pass	Pass	Pass	Pass	Pass
	C3.2	Pass	Pass	Fail	Pass	Pass	Partial	Fail	Fail	Pass	Partial	Pass	Pass	Pass	Pass	Pass	Pass
	C3.3	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Pass	Pass	Fail	Pass	Pass	Pass	Pass	Pass
	C3.4	Pass	Pass	Fail	Pass	Pass	Partial	Fail	Fail	Pass	Pass	Fail	Pass	Pass	Pass	Fail	Pass
	C3.5	Pass	Pass	Pass	Pass	Pass	Pass	Fail	Fail	Pass	Pass	Pass	Pass	Pass	Pass	Pass	Pass
	C3.6	Partial	Partial	Partial	Partial	Partial	Partial	Fail	Fail	Partial	Partial	Partial	Partial	Partial	Partial	Partial	Partial
QR Code	C4.1	NA	NA	NA	NA	NA	NA	NA	NA	Pass	Pass	Partial	Partial	NA	NA	NA	NA
	C4.2	NA	NA	NA	NA	NA	NA	NA	NA	Pass	Pass	Pass	Pass	NA	NA	NA	NA
	C4.3	NA	NA	NA	NA	NA	NA	NA	NA	Pass	Pass	Pass	Pass	NA	NA	NA	NA
	C4.4	NA	NA	NA	NA	NA	NA	NA	NA	Pass	Pass	Fail	Fail	NA	NA	NA	NA
	C4.5	NA	NA	NA	NA	NA	NA	NA	NA	Pass	Pass	Partial	Partial	NA	NA	NA	NA

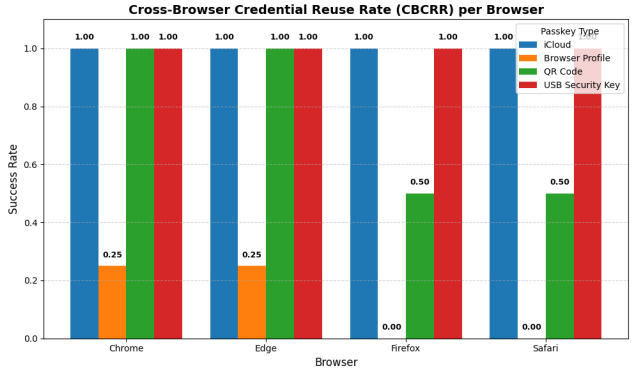


Fig. 3: Cross-Browser Credential Reuse Rate (CBCRR) per browser and passkey type.

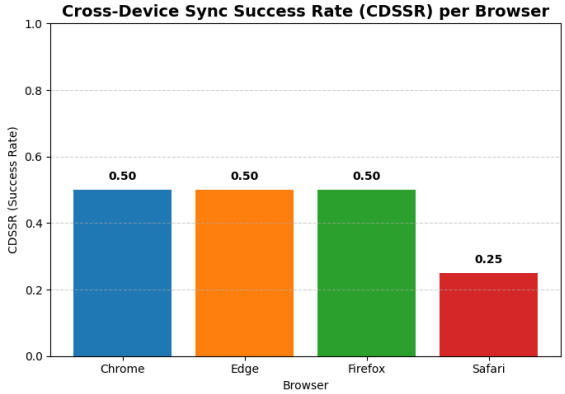


Fig. 4: Cross-Device Sync Success Rate (CDSSR).

ecosystem. Chrome, Edge, and Firefox each recorded two successful authentications out of 4 attempts, resulting in a CDSSR = 0.50. While Safari achieved only one successful authentication out of 4 attempts, corresponding to a CDSSR = 0.25. These results indicate that Chrome, Edge, and Firefox perform similarly, successfully authenticating in half of the cross-device attempts. In contrast, Safari demonstrates weaker cross-device interoperability, with success in only one-quarter of attempts. Half-success across Chrome, Edge, and Firefox reflects partial sync propagation via password managers; Safari's lower rate stems from stricter Keychain isolation.

Figure 5 shows the Roaming Authenticator Reuse Rate (RARR) across Chrome, Edge, Firefox, and Safari. RARR measures how often credentials stored on a roaming authenticator (e.g., a USB security key) can be reused for authentication across browsers. The results indicate that Chrome, Edge, and Firefox achieved full reuse (RARR = 1.00), successfully authenticating in every attempt with the roaming authenticator. In contrast, Safari failed to authenticate in any attempt (RARR = 0.00), demonstrating a lack of support for reusing USB security keys. These findings show that roaming authenticators are fully interoperable across Chrome, Edge, and Firefox, while Safari presents a clear interoperability gap for USB-based passkeys. Safari's zero reuse likely results from missing CTAP2 HID channel support for external authenticators.

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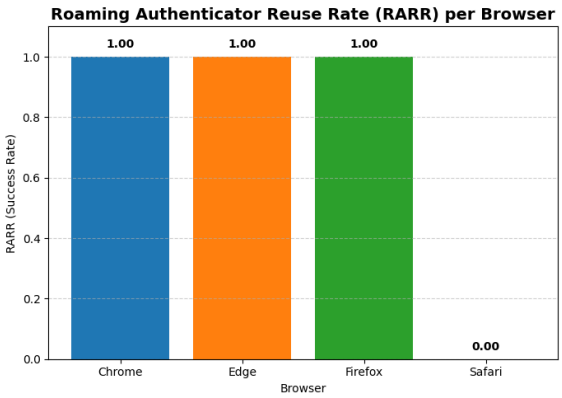


Fig. 5: Roaming Authenticator Reuse Rate (RARR).

C. Descriptive Case Studies

To illustrate the observed compliance and interoperability issues, we extracted three representative case studies from the experimental scenarios. Each case study highlights a specific deviation from expected behavior, supported by screenshots captured during testing.

Case Study 1 – Inconsistent Display of Alternative Passkey Options (C1.2/C3.2): During the registration process, Safari and Firefox immediately display the default passkey prompt along with an "Other Options" link, allowing users to select alternative authenticators (e.g., USB security keys or cross-device credentials). In contrast, Chrome and Edge initially present only the default passkey prompt without this link. The alternative options are revealed only after the user cancels the default prompt. This behavior introduces additional steps for Chrome and Edge users and may reduce discoverability of alternative passkeys, potentially impacting usability in multi-device or roaming authenticator scenarios.

Case Study 2 – Missing Consent Prompt (C1.4): Across all tested browsers, the registration process proceeded without displaying any consent prompt or informational message about the passkey being created. According to the WebAuthn specification, relying parties should obtain explicit user consent before creating a credential, ensuring that users understand the action being performed and its implications for future authentication.

The absence of such a prompt means users are not informed about where the credential will be stored or how it can be used. This omission reduces transparency and may negatively affect user trust, particularly for non-technical users.

Case Study 3 – Inconsistent Error Messaging Across Browsers (C1.6/C3.4): When users canceled a registration or authentication attempt, browsers displayed inconsistent error messages. Safari and Firefox returned the message: *"The request is not allowed by the user agent or the platform in the current context, possibly because the user denied permission."*

In contrast, Chrome and Edge displayed: *"The operation either timed out or was not allowed. See: https://www.w3.org/TR/webauthn-2/#sctn-privacy-considerations-client."*

Similarly, when attempting QR code–based registration on Safari, the same denial message was shown, whereas Chrome and Edge displayed: *"The operation could not be completed. Please try again."*

These inconsistencies reduce the clarity of the user experience and fail to provide guidance or fallback options, which contradicts WebAuthn's usability and security requirements for error handling (criteria C1.6 and C3.4).

VI. Discussion

The evaluation results reveal clear differences in passkey interoperability across browsers, passkey types, and authentication workflows. The CBCRR results (Figure 3) indicate that iCloud Keychain and USB security keys achieved full cross-browser reuse (CBCRR = 1.00), demonstrating strong adherence to the WebAuthn and FIDO2 specifications. In contrast, browser profile credentials entirely failed to support cross-browser authentication (CBCRR $\leq$ 0.25), confirming their isolated nature. QR code–based passkeys provided only partial interoperability (CBCRR = 0.50 in Firefox and Safari), suggesting that credential discovery and reuse remain inconsistent across browsers.

The CDSSR results (Figure 4) further demonstrate that Chrome, Edge, and Firefox successfully authenticated in half of the cross-device attempts, whereas Safari achieved only 25% success. These findings highlight the immaturity of cross-device synchronization, particularly for iCloud Keychain, compared with Google Password Manager and Windows Hello, which integrate more reliably with Chrome and Edge. This gap suggests that platform-specific implementations still vary in their support for multi-device credentials.

Similarly, the RARR results (Figure 5) show that Chrome, Edge, and Firefox consistently supported credential reuse with USB security keys (RARR = 1.00), while Safari failed to recognize roaming authenticators (RARR = 0.00). This absence of support for external authenticators in Safari contrasts sharply with the consistent CTAP2 support observed in the other browsers.

Collectively, these findings indicate that platform-synced passkeys (e.g., iCloud Keychain, Google Password Manager) and roaming authenticators currently offer the most reliable interoperability. In contrast, browser profile credentials and QR code–based workflows exhibit inconsistent implementation, limiting the seamless user experience envisioned by FIDO2 and WebAuthn. Similar challenges have been noted in prior studies on authentication system interoperability [9], [16], [20], but our work extends these findings through empirical evaluation across multiple browsers and passkey types.

From a usability and security perspective, the inconsistent handling of fallback mechanisms and error messaging (criteria C3.6 and C1.6) presents a significant barrier to adoption. While Chrome and Safari occasionally provided clearer feedback, Edge and Firefox often displayed generic or unclear error

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messages. The absence of a consent prompt in all browsers also raises concerns regarding transparency, as users receive little contextual information about passkey creation. These issues can undermine user trust, particularly for less technical users, and may discourage the transition to passwordless authentication.

Addressing these gaps requires concerted efforts from browser vendors and standards bodies. Standardizing fallback workflows, improving the discoverability of alternative authenticators, and ensuring that meaningful error messages and consent prompts are consistently presented are essential steps toward enhancing usability and interoperability. Furthermore, transparency in credential handling can strengthen user trust and promote the widespread adoption of passkeys. While this study focused on observable browser behaviors, future work should extend to protocol-level validation and include user studies to better understand how these inconsistencies affect real-world adoption.

VII. Study Limitations

This study focused on observable user interface behaviors using the WebAuthn.io testbed without packet-level analysis of FIDO2/CTAP2 exchanges. All experiments were conducted on macOS desktop browsers, so results may vary across other platforms. Additionally, no user studies were performed to assess perceived usability or trust. Future work will extend testing to multiple operating systems, include protocol-level tracing, and explore user perspectives on passkey adoption.

VIII. Conclusion and Future Work

This paper evaluated passkey implementations in Chrome, Safari, Edge, and Firefox for compliance with FIDO2/WebAuthn and practical interoperability. Tests with iCloud Keychain, browser profile credentials, QR code workflows, and USB security keys showed that while platform-synced and roaming credentials performed reliably, browser profile credentials failed, and cross-device synchronization was inconsistent, particularly in Safari. Future work will extend testing to additional platforms, include protocol-level validation, and conduct user studies to assess the impact of inconsistent behaviors on adoption.

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